

Supplement Material for Look Clearly Before Answering: Mitigating Hallucinations in LVLMs via Saliency-Driven Perceptual Realignment

This supplementary document is organized as follows:

section	content
Section 1	Additional experimental results (POPE, general LVLM benchmarks, and efficiency comparisons)
Section 2	Ablation studies with parameters
Section 3	Further studies building upon the discussion of Degradation of LVLMs’ Visual Awareness (Section 4 in manuscript)
Section 4	Additional case studies on open-ended generation tasks
Section 5	Additional experimental details (introduced benchmarks and alternative baselines)

1 More Experimental Results

In this section, we report the results on the POPE benchmark using Qwen (Table 3) and InternVL (Table 4) as base models, alongside performance comparisons on several general LVLM datasets in Table 5. Beyond task-specific accuracy, we also systematically measure the computational efficiency and max memory footprint of our method relative to existing SOTA baselines in Table 1. Finally, we rep the performance on lager-scale LVLM in Table 2.

Additional Results on POPE. As demonstrated in Table 3 and Table 4, our proposed SDPR consistently yields superior performance on the POPE benchmark. Using Qwen-VL as the backbone, SDPR achieves a significant average improvement of 1.69%, 1.58%, and 3.02% in accuracy, along with gains of 2.08%, 1.79%, and 3.51% in F1-score across the three dataset splits. Notably, our method outperforms state-of-the-art competitors, surpassing VCD, M3ID, and ONLY by 0.7%, 1.6%, and 0.86% respectively. Similarly, when employing InternVL as the backbone, SDPR continues to deliver robust results. Compared to the baseline, it achieves substantial accuracy enhancements of 5.10%, 5.13%, and 5.32% across the three data subsets, alongside F1-score increases of 4.4%, 4.38%, and 4.62%, ultimately attaining state-of-the-art performance. These consistent findings across diverse model architectures firmly substantiate the exceptional generalizability of our proposed SDPR in mitigating hallucinations.

Performance on General LVLM Dataset. Additionally, we assessed the performance of SDPR using general LVLM datasets (MME_{RealWorld}, MMVP, MMMU, MMMU_{pro}, and ScienceQA). As demonstrated by Table 5, our proposed SDPR consistently achieves the highest overall average score of 42.7, surpassing the baseline and all other competitive approaches across the evaluated architectures.

When analyzing datasets focused on *visual perception* and specific scientific reasoning, SDPR exhibits remarkable superiority. On the MME_{RealWorld} benchmark, it secures the absolute top performance across the LLaVA-V1.5-7B, Qwen2.5-VL-7B, and InternVL2.5-8B base models. Similarly, on the MMVP dataset, SDPR attains the

highest scores for both the Qwen and InternVL architectures while maintaining a strong second-best position on LLaVA. For scientific image reasoning tasks evaluated by ScienceQA, the method sets new peak records on the LLaVA and InternVL models and delivers highly competitive second-best results on Qwen.

Furthermore, the robustness of SDPR is distinctly evident in *complex multidisciplinary multimodal understanding* scenarios. On the standard MMMU benchmark, our method achieves the highest scores across all evaluated models, registering significant improvements over previous techniques. When subjected to the more challenging MMMU_{pro} variant, SDPR continues to demonstrate dominant capabilities by securing the best performance on the Qwen architecture alongside robust second-best results on both the LLaVA and InternVL setups. This consistent achievement of top-tier results across diverse dataset categories highlights the comprehensive effectiveness of SDPR in advancing general large vision-language model capabilities.

Efficiency Comparison. As summarized in Table 1, we evaluate the computational efficiency and memory footprint of SDPR against several state-of-the-art baselines. As a contrastive-based approach, SDPR incurs significantly lower GPU memory overhead compared to typical contrastive decoding methods such as VCD, M3ID, AvisC, and MoD. Furthermore, by eliminating the need for redundant forward passes, our method achieves a higher inference speed than these baselines. Although attention-based intervention methods (e.g., VAR and VAF) may offer slightly higher raw efficiency, SDPR substantially outperforms them in terms of generation accuracy. Ultimately, our method delivers state-of-the-art performance with negligible computational increment, significantly bolstering the faithfulness and factuality of the generated content while effectively mitigating visual hallucinations.

Performance on Larger-Scale LVLMs. We extend our evaluation to the 13B variant of the LLaVA-1.5 model to assess the scalability of our approach.

The detailed results in Table 2 indicate that SDPR continues to exhibit exceptional generalization and stability on models with larger parameter scales. In real-world visual perception evaluations (e.g., POPE, MME, and MME_{RW}), SDPR delivers superior performance: compared to the baseline models, it yields absolute performance gains of 6.0 and 4.07 on the POPE and MME_{RW} datasets, respectively, comprehensively surpassing the majority of the compared methods. Additionally, SDPR demonstrates remarkable competitiveness in complex reasoning tasks, boosting metrics on MMMU and ScienceQA by 7.5 and 1.8, respectively, and achieving optimal results that existing baselines fail to reach. These results demonstrate that SDPR can effectively accommodate hallucination mitigation demands across varying parameter scales and complex scenarios. It maintains high efficacy within large-scale models, providing robust support for the safe and trustworthy deployment of LVLMs.

Table 1: Efficiency Comparison. The average peak GPU memory and inference latency per token are presented.

Method	GPU Mem ↓	Time (ms) ↓
Vanilla	16,820 ×1.00	23.41 ×1.00
VCD	17,186 ×1.02	45.03 ×1.92
M3ID	17,186 ×1.02	44.03 ×1.88
VAR	16,825 ×1.00	24.05 ×1.03
VAF	16,825 ×1.00	23.41 ×1.00
ONLY	16,840 ×1.00	25.37 ×1.08
AvisC	19,036 ×1.13	27.00 ×1.15
MoD	19,036 ×1.13	45.15 ×1.93
SDPR	16,840 ×1.00	24.64 ×1.05

Table 2: Performance on Larger-scale LVLm, higher is better. The best performance is shown in bold, and the second-best is underlined. MME_{RW} denotes the MME RealWorld dataset, and SQA_{img} refers to the visual QA split of ScienceQA.

<i>LLaVA-V1.5-13B</i>	<i>POPE</i>	<i>MME</i>	<i>MME_{RW}</i>	<i>MMNVP</i>	<i>MMMU</i>	<i>SQA_{img}</i>
Vanilla	79.2	1652	33.35	32.7	24.8	69.4
VCD <i>CVPR'24</i>	81.4	1690	34.24	34.0	28.6	70.0
M3ID <i>CVPR'24</i>	81.6	1720	33.45	33.3	28.1	70.0
VAR <i>ICLR'25</i>	80.1	1711	30.38	28.7	27.3	67.4
VAF <i>CVPR'25</i>	80.3	1690	34.55	31.3	28.2	69.0
ONLY <i>ICCV'25</i>	<u>84.9</u>	1812	<u>37.10</u>	33.3	29.4	69.6
AvisC <i>ACL'25</i>	84.3	1804	36.52	31.3	27.1	69.5
MoD <i>ACL'25</i>	84.6	1844	36.91	35.3	30.5	70.0
SDPR <i>ours</i>	85.2	<u>1834</u>	37.42	<u>34.0</u>	32.3	71.2

2 Ablation Study on Parameter of SDPR

In this section, we conduct a comprehensive ablation study to investigate the sensitivity of SDPR to key hyper-parameters across its three constituent modules: SDAR (perception), SDCA (memory), and PCD (prediction). Specifically, we first evaluate the perception enhancement module SDAR by analyzing the number of idle heads, the number of redundant tokens removed, and the specific layers designated for redistribution. Subsequently, for the memory module, we examine the layer selection to compute query-specific saliency weights within SDCA, as well as the range of layers subject to KV cache constraints. Finally, we assess the prediction module by exploring the impact of the adaptive decoding threshold γ of the PCD framework. This systematic analysis aims to reveal the contribution of each design choice to the overall performance-efficiency trade-off.

2.1 Study on SDAR

Hyper-Parameters in SDAR. To investigate the influence of hyper-parameters within the SDAR module, we first analyze the capacity of the idle head set $\mathcal{H}_{idle}$, which is determined by selecting the most inactive heads based on entropy. We vary the number of idle heads from 4 to 20, with results summarized in Table 6. Our empirical findings indicate that treating half of the heads (*i.e.* 16 heads)

as inactive yields the peak MME accuracy. This suggests that re-activating 16 heads effectively enhances the model’s visual perception capabilities. Subsequently, we examine the optimal number of salient tokens to be redistributed per head. As shown in Table 6, we evaluate the removal of 3 to 20 tokens. The model performance reaches its zenith when 5 tokens are redistributed; beyond this threshold, a further increase in the number of reassigned tokens leads to a degradation in visual recognition accuracy. Consequently, reallocating the top-5 sink tokens serves as an optimal configuration to balance comprehensive performance and perceptual fidelity.

Layers of Applying SDAR. We conduct experiments to investigate the optimal stage for integrating SDAR, specifically identifying where intervention against attention hijacking is most effective. Prior research indicates that mainstream LVLms undergo intensive vision-language interactions within Layers 5 to 15 [1]. Guided by this observation, we evaluate the performance across different intervention windows. Specifically, starting from Layer 0, we define five experimental groups, each covering a span of 10 consecutive layers to assess their collective impact.

As reported in Table 7, the configuration targeting Layers 5-15 achieves the superior performance. This result validates that during the stages of frequent cross-modal interaction, our SDAR effectively reactivates idle attention heads and enhances the model’s visual perception capabilities by mitigating premature attention hijack.

Furthermore, we conduct a more granular investigation of the performance within the 5-15 layer range by partitioning it into two sub-intervals. As illustrated in Table 7, targeting only Layers 5-10 or 10-15 independently does not yield superior results compared to the full 5-15 span. This observation suggests that vision-language interactions are densely distributed across the entire 5-15 layer region rather than being localized within a smaller subset. Consequently, applying our SDAR-based perceptual enhancement across this integral interaction zone is essential to fully unleash the model’s visual perception capacity and achieve optimal performance.

2.2 Study on SDCA.

Layers of Generating Query-Specific Saliency Weight. In SDCA, we introduce the Query-Specific Saliency Weight to constrain the visual semantic regions within the KV cache. We conduct an analysis to determine the optimal layer for extracting features to guide the KV cache during generation. To evaluate this comprehensively, we report the CHAIR Joint Value ($CHAIR_{joint} = CHAIR_S + CHAIR_I$, lower is better), which integrates hallucination rates at both sentence and instance levels. As shown in Figure 1, the $CHAIR_{joint}$ remains high when using features earlier than Layer 5, indicating that low-level representations are unsuitable as visual memory cues. From Layer 5 to 15, a stage of intensive vision-language interaction, $CHAIR_{joint}$ decreases consistently and reaches its minimum of 10.6 at $L = 15$. While further depth yields secondary local minima at Layer 23 (10.9) and Layer 31 (10.8), these gains are marginal. Considering the entire generation trajectory, we select $L = 15$ as the optimal layer for generating the query-specific saliency weights in SDCA.

Furthermore, we conduct a more granular investigation of the performance within the 5-15 layer range by partitioning it into two sub-intervals. As illustrated in Table 7, targeting only Layers 5-10 or

Table 3: POPE performance comparison on different benchmarks using Qwen2.5-VL-7B.

POPE Qwen2.5-VL		Random		Popular		Adversarial		Average	
		Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑
COCO	Vanilla	85.666	83.687	84.933	83.071	83.266	81.516	84.622	82.758
	VCD <i>CVPR'24</i>	87.433	85.979	85.767	84.295	83.633	82.161	85.611	84.145
	M3ID <i>CVPR'24</i>	85.666	83.625	85.167	83.302	83.300	81.668	84.711	82.865
	VAR <i>ICLR'25</i>	85.866	83.816	85.000	83.031	83.366	81.593	84.744	82.813
	VAF <i>CVPR'25</i>	87.067	85.587	85.267	83.653	83.433	82.180	85.256	83.807
	ONLY <i>ICCV'25</i>	86.700	84.858	85.900	84.127	83.766	82.089	85.455	83.691
	AvisC <i>ACL'25</i>	82.900	80.000	82.800	80.100	81.200	78.500	82.300	79.533
	MoD <i>ACL'25</i>	86.000	84.100	85.900	84.127	84.133	82.487	85.344	83.571
	SDPR <i>ours</i>	87.533	85.993	86.633	85.121	84.767	83.387	86.311	84.834
A-OKVQA	Vanilla	86.600	85.264	86.267	85.094	81.333	80.729	84.733	83.696
	VCD <i>CVPR'24</i>	88.133	87.258	87.233	86.078	81.433	81.377	85.600	84.904
	M3ID <i>CVPR'24</i>	87.233	86.027	86.767	85.526	81.867	81.305	85.289	84.286
	VAR <i>ICLR'25</i>	86.400	84.988	85.667	84.249	80.833	80.013	84.300	83.083
	VAF <i>CVPR'25</i>	87.700	86.844	87.033	85.900	82.167	82.148	85.633	84.964
	ONLY <i>ICCV'25</i>	87.967	86.820	87.567	86.430	82.433	81.858	85.989	85.036
	AvisC <i>ACL'25</i>	84.700	83.000	83.900	82.400	78.100	77.300	82.233	80.900
	MoD <i>ACL'25</i>	87.533	86.330	87.600	86.492	81.900	81.243	85.678	84.688
	SDPR <i>ours</i>	88.367	87.396	88.033	87.072	82.533	81.997	86.311	85.488
GQA	Vanilla	84.833	83.650	80.333	79.542	80.066	79.797	81.744	80.996
	VCD <i>CVPR'24</i>	85.900	85.058	81.067	81.079	79.933	79.960	82.300	82.032
	M3ID <i>CVPR'24</i>	85.833	84.805	81.833	81.291	80.333	80.081	82.666	82.059
	VAR <i>ICLR'25</i>	85.100	83.487	82.167	80.954	79.900	78.952	82.389	81.131
	VAF <i>CVPR'25</i>	84.533	83.889	80.467	80.388	79.400	79.671	81.467	81.316
	ONLY <i>ICCV'25</i>	87.233	86.297	83.600	83.058	82.033	81.710	84.289	83.688
	AvisC <i>ACL'25</i>	80.500	77.800	74.200	72.300	75.500	78.100	76.733	76.067
	MoD <i>ACL'25</i>	87.300	86.436	83.600	83.162	81.733	81.511	84.211	83.703
	SDPR <i>ours</i>	88.033	87.337	84.174	83.901	82.088	82.270	84.765	84.503

Table 4: POPE performance comparison on different benchmarks using InternVL2.5-8B.

POPE InternVL2.5		Random		Popular		Adversarial		Average	
		Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑	Accuracy ↑	F1-Score ↑
COCO	Vanilla	79.900	81.173	76.867	78.841	73.666	76.627	76.811	78.880
	VCD <i>CVPR'24</i>	85.000	85.294	80.433	81.753	77.733	79.634	81.055	82.227
	M3ID <i>CVPR'24</i>	85.355	85.582	80.403	81.639	77.139	79.918	80.966	82.380
	VAR <i>ICLR'25</i>	77.167	79.185	74.066	77.023	71.900	75.429	74.378	77.212
	VAF <i>CVPR'25</i>	79.133	80.655	76.233	78.215	72.600	76.118	75.989	78.329
	ONLY <i>ICCV'25</i>	85.667	86.217	81.700	82.784	76.000	78.584	81.122	82.528
	SDPR <i>ours</i>	86.233	86.603	82.633	83.785	76.866	79.443	81.911	83.277
A-OKVQA	Vanilla	79.100	81.017	76.133	78.879	70.667	75.085	75.300	78.327
	VCD <i>CVPR'24</i>	83.567	84.684	79.833	81.869	72.500	76.806	78.633	81.120
	M3ID <i>CVPR'24</i>	83.713	84.768	80.053	81.883	72.122	76.614	78.629	81.088
	VAR <i>ICLR'25</i>	77.767	79.939	72.133	76.330	67.533	73.182	72.478	76.484
	VAF <i>CVPR'25</i>	77.867	79.987	74.933	78.075	68.833	74.164	73.878	77.409
	ONLY <i>ICCV'25</i>	85.600	86.525	80.567	82.338	74.233	78.281	80.133	82.381
	SDPR <i>ours</i>	86.033	86.959	80.867	82.773	74.400	78.401	80.433	82.711
GAQ	Vanilla	78.567	80.000	74.466	77.510	69.633	73.934	74.222	77.148
	VCD <i>CVPR'24</i>	82.833	83.881	79.166	80.872	73.566	77.166	78.522	80.640
	M3ID <i>CVPR'24</i>	82.491	83.482	78.659	80.827	73.311	77.346	78.154	80.552
	VAR <i>ICLR'25</i>	76.400	79.028	70.533	74.943	67.100	72.681	71.344	75.551
	VAF <i>CVPR'25</i>	76.233	78.697	73.533	76.591	69.500	74.319	73.089	76.536
	ONLY <i>ICCV'25</i>	84.300	85.286	79.467	81.333	74.100	77.806	79.289	81.475
	SDPR <i>ours</i>	84.566	85.380	79.867	81.719	74.200	78.192	79.544	81.764

Table 5: General LVLm benchmark performance comparison, higher is better. The best performance is shown in bold, and the second-best is underlined. MME_{RW} denotes the MME RealWorld dataset, and SQA_{img} refers to the visual QA split of ScienceQA.

	<i>LLaVA-V1.5-7B</i>					<i>Qwen2.5-VL-7B</i>					<i>InternVL2.5-8B</i>					Avg.
	MME_{RW}	$MMVP$	$MMMU$	$MMMU_{pro}$	SQA_{img}	MME_{RW}	$MMVP$	$MMMU$	$MMMU_{pro}$	SQA_{img}	MME_{RW}	$MMVP$	$MMMU$	$MMMU_{pro}$	SQA_{img}	
Vanilla	30.4	27.3	26.1	25.3	65.9	22.6	29.3	25.1	22.1	60.2	31.5	43.0	33.7	36.7	96.7	38.4
VCD <i>CVPR'24</i>	34.0	29.3	25.2	26.2	65.2	23.7	24.0	26.8	23.3	62.0	32.8	39.3	36.1	38.0	97.0	38.9
M3ID <i>CVPR'24</i>	33.2	30.0	26.1	26.5	65.5	24.4	22.0	27.8	23.6	63.4	32.3	43.3	40.7	38.4	97.1	39.6
VAR <i>ICLR'25</i>	28.7	29.3	25.2	24.4	64.0	19.5	22.7	9.2	22.8	58.8	34.1	42.0	41.3	35.7	97.2	37.0
VAF <i>CVPR'25</i>	33.4	26.6	26.1	25.0	65.4	23.6	22.7	27.6	23.7	61.8	32.7	40.0	34.6	37.1	96.7	38.5
ONLY <i>ICCV'25</i>	33.5	28.7	23.2	28.0	<u>67.1</u>	<u>25.0</u>	28.0	32.4	<u>26.9</u>	67.9	32.5	43.3	35.2	40.1	97.3	40.6
AvisC <i>ACL'25</i>	33.9	22.0	<u>27.0</u>	25.4	65.1	23.1	28.0	31.8	24.6	67.0	35.0	42.0	40.6	39.4	97.3	40.1
MoD <i>ACL'25</i>	<u>34.5</u>	26.0	25.2	26.8	65.2	24.0	<u>28.7</u>	<u>33.8</u>	26.4	67.1	<u>35.3</u>	38.0	<u>41.8</u>	40.5	<u>97.5</u>	<u>40.7</u>
SDPR <i>ours</i>	34.6	<u>29.3</u>	31.0	<u>27.2</u>	67.6	26.1	29.3	36.2	27.0	<u>67.5</u>	37.8	46.0	43.0	<u>40.2</u>	97.7	42.7

Table 6: Ablation study on hyper-parameters of the SDAR.

Heads	Tokens	MME $\uparrow$	Coarse. $\uparrow$
4	5	1840	640
8	5	1835	635
12	5	1850	635
16	5	1856	627
20	5	1843	628
16	3	1840	630
16	5	1856	627
16	10	1849	630
16	15	1843	625
16	20	1834	620

Table 7: Ablation study on the selection of layer ranges for SDAR.

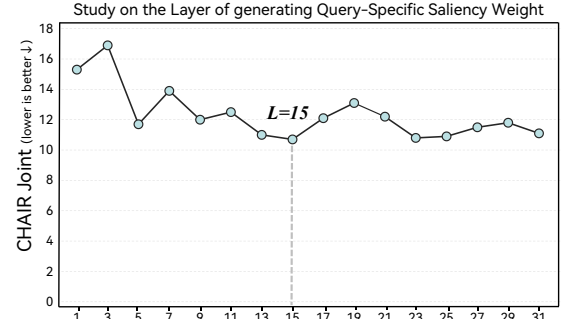
Layers	0 to 10	5 to 15	10 to 20	15 to 25	20 to 30	5 to 10	10 to 15
MME $\uparrow$	1817	1856	1782	1785	1791	1843	1824

10-15 independently does not yield superior results compared to the full 5-15 span. This observation suggests that vision-language interactions are densely distributed across the entire 5-15 layer region rather than being localized within a smaller subset. Consequently, applying our SDCA-based perceptual enhancement across this integral interaction zone is essential to fully unleash the model’s visual perception capacity and achieve optimal performance.

Layers of Applying SDCA. We conduct ablation studies to investigate the optimal stage for integrating SDCA. Following an experimental setup similar to SDAR, we establish five groups of experiments using Layer 0 as the baseline, with each group evaluating the collective performance of 10 consecutive layers. As demonstrated in Table 8, selecting layers 5 to 15 yields the optimal performance. This suggests that within the layers characterized by intensive cross-modal interactions, our SDCA method effectively enhances the visual memory of newly generated tokens during the inference phase by injecting query-related saliency regions into the

Table 8: Ablation study on the selection of layer ranges for SDCA.

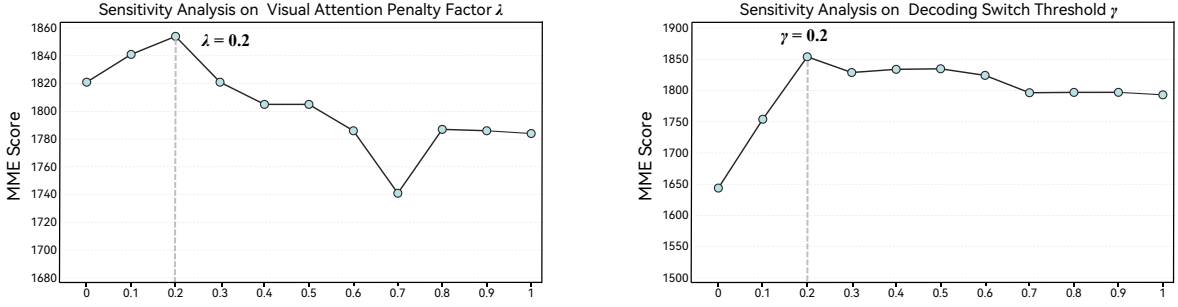
Layers	0 to 10	5 to 15	10 to 20	15 to 25	20 to 30
$CHAIR_S \downarrow$	11.6	7.2	10.2	9.8	11.2
$CHAIR_I \downarrow$	5.8	3.4	5.9	5.4	6.0
$CHAIR_{Joint} \downarrow$	17.4	10.6	16.1	15.2	17.2

Figure 1: Left: Layer of generating Query-Specific Saliency Weight S_{query} .

KV cache. Consequently, this improves the faithfulness of the generated sequences and significantly mitigates visual hallucinations.

2.3 Study on PCD.

Visual Attention Penalty Factor (λ). Regarding the Prior-constrained Contrastive Decoding, we first analyze the influence of the Visual Attention Penalty Factor (λ) and provide a sensitivity analysis on the Decoding Switch Threshold. As illustrated in left part of Figure 2, the LVLm’s overall accuracy improves as the visual penalty factor λ increases from 0 to 0.2, followed by a consistent decline for values exceeding 0.2. We argue that an excessively high penalty factor provides insufficient suppression of visual information in negative samples, thereby failing to accurately represent predictive bias. Conversely, a penalty factor that is too low creates an

Figure 2: Left: Impact of the visual attention penalty factor λ .Right: Sensitivity analysis of decoding mode switch threshold γ .

excessive gap between positive and negative samples, potentially leading to misalignment in the prediction space. Therefore, $\lambda = 0.2$ is identified as the optimal configuration.

Decoding Switch Threshold (γ). We further evaluate the influence of the Decoding Switch Threshold (γ). As depicted in Figure 2, a significant performance gain is observed as γ increases from 0 to 0.2, with the model reaching its peak accuracy at $\gamma = 0.2$. This indicates that at this threshold, the LLaVA-based architecture achieves its optimal performance, successfully balancing the transition between the contrast and collaboration decoding phases. Beyond this point, further increases in γ result in performance fluctuations and an eventual downward trend. Based on these empirical findings, we identify $\gamma = 0.2$ as the optimal threshold for the decoding switch mechanism. Beside, we set $\gamma = 0.4$ for Qwen and InternVL.

Decoding Multiplier Parameter (α_1, α_2). We conducted experiments to analyze the impact of the positive and negative logit multipliers, which regulate the influence of the prejudiced branch. As shown in Figure 3, the default configuration ($\alpha_1 = 3, \alpha_2 = 1$) consistently yields the best performance.

3 Additional Study building upon Section 4 : Degradation of LLMs' Visual Awareness

We further conduct studies building upon the discussion of Section 4: Degradation of LLMs' Visual Awareness. We first provide a comparative study highlighting the practical limitations of existing sink-localization methods. After that, we conduct both qualitative and quantitative analyses to investigate the attention hijack phenomenon and the role of hijacking attention release. These findings collectively substantiate the motivation behind our proposed SDAR, demonstrating its necessity and effectiveness.

3.1 Study on Visual Attention Sink Localization

The Dilemma of Sink Localization. Although the preceding content validates that removing sinks effectively liberates suppressed visual perception, a critical challenge remains: **how to accurately localize these sink tokens?** VAR [2] introduces a solution by identifying anomalous activation peaks: Given a hidden state $x \in \mathbb{R}^D$, the RMS-norm is calculated as

$$\phi(x) = \max_{d \in \mathbb{D}_{\text{sink}}} |x[d]| / \sqrt{\frac{1}{D} \sum_{d=1}^D |x[d]|^2}, \quad (1)$$

where tokens exceeding a fixed threshold τ are identified as sinks: $S = \{x \mid \phi(x) > \tau\}$, where tokens surpassing a predefined threshold $\tau = 20$ are classified as sinks. As illustrated in Figure 4, while VAR successfully identifies certain meaningless tokens marked with **red** bounding boxes, it overlooks several other heightened yet semantically-void tokens indicated by **yellow** boxes.

To further assess the impact of these overlooked tokens, we apply the weight redistribution scheme of VAR. However, as shown in Figure 5 (c), the sink pattern persists due to these residual tokens. Further analysis of the RMS-norm distribution in Figure 5 (h) reveals the cause: the unremoved sink tokens (yellow box) fall within the normal range, exhibiting no statistical abnormality. This exposes a fundamental limitation of hidden state-based detection: RMS-norm primarily reflects FFN amplification, thereby failing to capture tokens that function as sinks purely due to attention allocation anomalies. Conversely, by extending *Inspiration Study 1* with an attention-based selection strategy, we identify 10 prominent sink tokens whose removal effectively liberates the suppressed semantic attention, as depicted in Figure 5 (d) and Figure 5 (e).

3.2 Study on Visual Attention Sink Removal.

To demonstrate the ubiquity of attention sinks and the generalizability of our mitigation strategies, we conduct both qualitative and quantitative analyses in this subsection. these evaluations serve to further substantiate the observations presented in Section 4 of the manuscript.

Qualitative Analysis. We visualize the attention patterns across representative samples to illustrate the prevalence of the sink phenomenon and the efficacy of attention-based sink removal. The result can be indicated in Figure 7. Red bounding boxes highlight the most prominent tokens, the majority of which are identified as attention sinks. We observe that the gradual mitigation of these sinks effectively uncovers latent visual perception capabilities that were previously suppressed. However, a distinct inflection point occurs when more than 12 tokens are removed, where the model's performance begins to degrade, suggesting that semantically meaningful tokens are being erroneously discarded. This further underscores a critical trade-off: while the removal of prominent sink tokens is essential for alleviating visual suppression, it is equally imperative to safeguard the semantic integrity of informative content to avoid unintended key information loss.

Quantitative Analysis. To investigate the performance gains from eliminating sink tokens, we conduct a quantitative analysis.

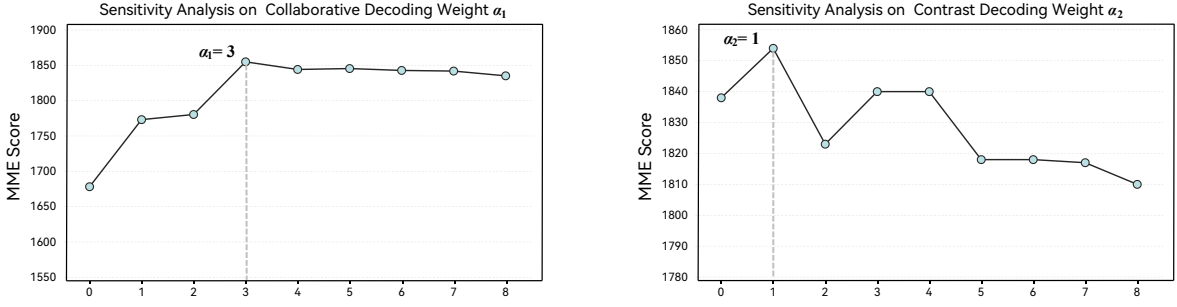


Figure 3: Left: Impact of the collaborative decoding weight α_1 . Right: Impact of the contrast decoding weight α_2 .

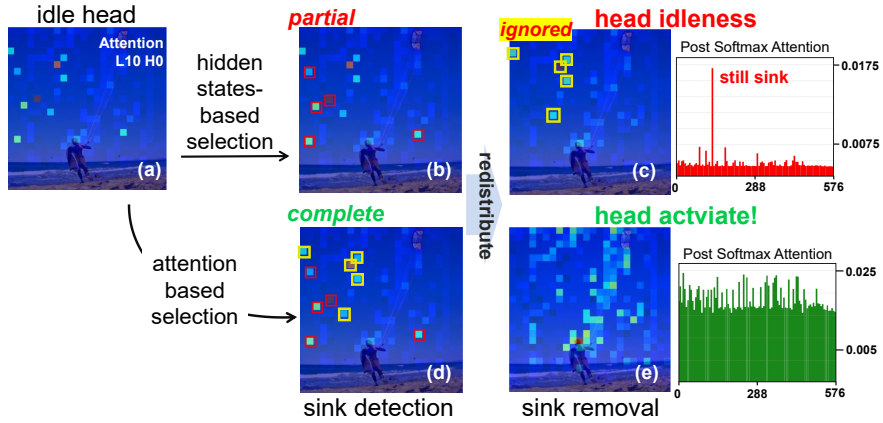


Figure 4: Different sink tokens selection. Two different visual sink selection strategies based on hidden states (VAR) and attention weight (ours SDPR) are presented.

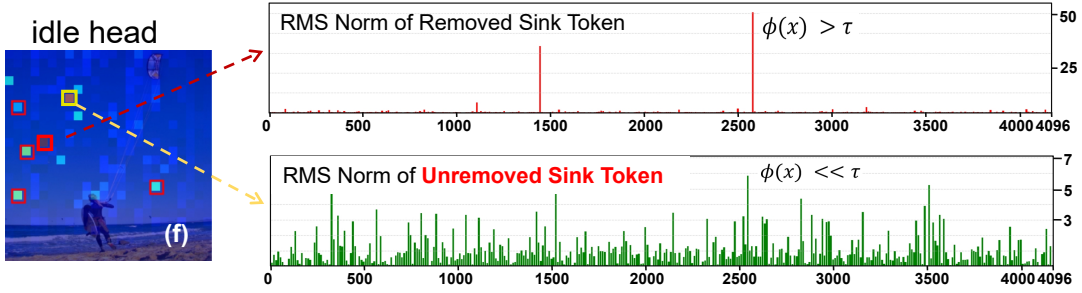


Figure 5: Failure analysis of RMS Norm-based detection. Tokens identified by VAR are highlighted with red bounding boxes, while those missed are marked in yellow. Notably, the undetected sink tokens fail to manifest the anticipated surge in activation magnitudes. This indicates that activation based methods cannot fully detect phantom tokens, particularly those that do not exhibit extreme high activation values.

Specifically, we iteratively remove the most salient tokens based on attention weights. As reported in Figure 6, we observe a progressive improvement in the model’s overall performance as the number of removed salient tokens increases. The performance reaches its peak when 10 tokens are removed, which aligns with the qualitative behavior illustrated in Figure 7. This trend suggests that purging the most dominant tokens successfully liberates the hijacked attention, thereby revitalizing the model’s perceptual sensitivity.

However, a performance inflection occurs beyond the removal of 12 tokens. We observe a subsequent decline, with performance falling below the original baseline when 15 tokens are eliminated. These results indicate that while removing sink tokens can unbind the perceptual capacity of attention heads, a purely semantic-agnostic removal strategy inevitably risks compromising essential semantic information. Consequently, this necessitates a more nuanced approach that can **selectively mitigate the adverse effects**

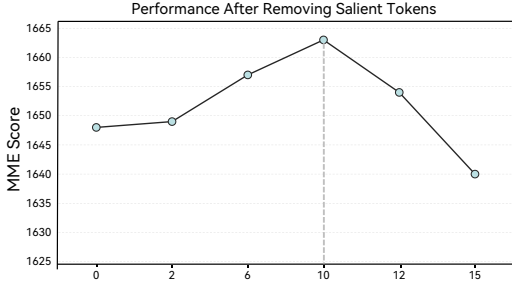


Figure 6: Performance After Removing Salient Tokens.

of sink tokens while preserving crucial semantic integrity, ultimately fully unlocking the visual perception capabilities of LVLm.

4 More Case Studies

We provide additional case studies in Figure 8. As illustrated, LLaVA demonstrates a deficiency in attending to critical visual cues during the initial perception phase. This limitation stems from the observation that sink tokens sequester a disproportionate amount of visual attention, thereby hindering the model’s ability to achieve accurate scene understanding. In contrast, our proposed SDPR exhibits more robust feature representation by effectively liberating the attention resources typically dominated by sink tokens. Furthermore, during the memory-retention stage, LLaVA suffers from a degradation of visual information characterized by significant serialization effects, which frequently leads to the generation of hallucinated content in long-form responses. Conversely, by accurately preserving visually salient information, SDPR generates more grounded and faithful responses, significantly mitigating the hallucination.

5 More Experimental Details

5.1 Details of Benchmarks

We conduct extensive experiments on the following benchmarks:

(1) **POPE** [3] is a widely recognized benchmark for evaluating object hallucinations in LVLms, which queries models using yes-or-no questions such as, “Is there a {object} in the image?”. The images in this benchmark are derived from three datasets, MSCOCO [4], A-OKVQA[5], and GQA [6], and the benchmark is structured into three distinct sub-datasets, namely random, popular, and adversarial, based on the negative sampling strategy employed. For each dataset configuration, there are 6 questions per image, resulting in a total of 3,000 test instances. To measure the comprehensive performance of each method, we calculate the Accuracy (ACC) and F1-score (F1), averaging the results across the three sub-datasets to obtain a holistic understanding of model reliability.

(2) **CHAIR** [7] is an evaluation benchmark designed to assess object hallucinations in LVLms through the task of image captioning. This benchmark utilizes a subset of 500 images randomly selected from the MSCOCO validation set, prompting models to generate descriptive responses for each. Performance is quantified using two primary metrics: $CHAIR_I$ (C_I , instance-level) and $CHAIR_S$ (C_S , sentence-level). Specifically, $CHAIR_I$ measures the proportion of

hallucinated objects relative to all mentioned objects:

$$CHAIR_I = \frac{\# \text{ hallucinated objects}}{\# \text{ all objects mentioned}} \quad (2)$$

Similarly, $CHAIR_S$ calculates the ratio of sentences containing at least one hallucinated object to the total number of sentences generated:

$$CHAIR_S = \frac{\# \text{ sentences with hallucinated object}}{\# \text{ all sentences}} \quad (3)$$

For both metrics, lower scores indicate a higher degree of factual reliability and a lower frequency of hallucination. While CHAIR indices focus on the error rate by quantifying the proportion of hallucinated entities, precision is introduced to characterize the correctness of the model’s output. It represents the proportion of correctly identified objects when the model asserts their presence, thereby serving as a vital measure of the model’s ability to resist imaginary hallucinations stemming from language priors.

(3) **MME** [8] is a holistic benchmark that systematically evaluates the perception and cognition capabilities of LVLms across 4 categories of tasks (Coarse-grained, Fine-grained, OCR, and Reasoning tasks) and 14 diverse subtasks. The perception category involves object-level evaluations such as existence and count, attribute-level assessments including position and color, and specific recognition tasks like celebrities and landmarks. The cognition category focuses on higher-order abilities, encompassing commonsense reasoning, numerical calculation, and code reasoning. MME uniquely adopts a binary-choice paradigm requiring correct answers for Yes/No pairs to eliminate the impact of random guessing.

Performance is quantified through a total score that aggregates two distinct metrics: Accuracy and Accuracy+; Accuracy is calculated based on the correctness of individual question responses. Accuracy+ is a more stringent measure awarded only when the model correctly answers both questions within a specific Yes/No pair for a single image. The final score for each subtask is calculated by summing the Accuracy and Accuracy+ results, and the overall benchmark performance is represented by the cumulative total across all evaluated categories. In this work, we evaluate our method across the *full MME dataset* to provide a comprehensive analysis of its performance in both perception and reasoning scenarios.

(4) **MMVP** [9] is designed to expose fine-grained visual deficits in LVLms by utilizing 150 curated CLIP-blind image pairs. These pairs feature subtle visual differences that standard vision encoders often fail to distinguish, paired with binary-choice questions to evaluate precise perception. To ensure a robust assessment and eliminate the impact of random guessing, the benchmark implements a stringent scoring protocol where a response is considered correct only if the model accurately answers the questions for both images within a pair.

(5) **MMMU** [10] is a massive multi-discipline multimodal understanding benchmark designed to evaluate LVLms on tasks requiring college-level knowledge and deliberate reasoning. It comprises 11,500 meticulously curated questions spanning 30 technical subjects across six core disciplines: Art & Design, Business, Science, Health & Medicine, Humanities & Social Science, and Tech & Engineering. Unlike benchmarks focused on common daily scenes,

MMU incorporates complex visual inputs such as specialized diagrams, chemical structures, and mathematical charts sourced from academic textbooks. This shift from basic perception to expert-level cognition provides a rigorous framework for assessing a model's multidisciplinary integration and advanced problem-solving capabilities in professional and technical domains.

(6) **MMU Pro** [11] is an advanced evaluation benchmark designed to rigorously assess the multimodal reasoning capabilities of LVLMs by mitigating the influence of text-only biases. Building upon the original MMU dataset, MMU Pro increases the evaluation difficulty by converting standard text-based multiple-choice options into vision-only inputs or by requiring models to generate open-ended responses. This paradigm forces LVLMs to rely on profound multidisciplinary understanding and authentic visual perception rather than exploiting textual shortcuts or language priors. The benchmark evaluates expert-level cognition across various technical and scientific disciplines. Performance is quantified using Accuracy, providing a stringent measure of a model's true cognitive proficiency and robustness against superficial guessing strategies in professional domains.

(7) **MME RealWorld** [12] is a comprehensive benchmark constructed to assess the perception and reasoning capabilities of LVLMs in challenging, high-resolution real-world scenarios. In contrast to traditional datasets that often rely on standard, web-crawled images, MME RealWorld features a massive collection of high-quality images captured in complex physical environments. It spans numerous practical applications, including autonomous driving, remote sensing, and defect detection. The benchmark challenges models to extract fine-grained local details from visually cluttered and dynamic scenes, bridging the gap between academic evaluation and practical deployment. Performance is primarily quantified through Accuracy, offering critical insights into the robustness and reliability of models when confronted with the unpredictability of everyday settings.

(8) **ScienceQA** [13] is a large-scale, multimodal educational benchmark designed to evaluate LVLMs on complex scientific reasoning tasks. The dataset comprises over 21,000 meticulously annotated multiple-choice questions spanning three primary disciplines: natural science, social science, and language science. A distinguishing feature of ScienceQA is its rich incorporation of diverse visual inputs—such as scientific diagrams, mathematical charts, and physical illustrations—coupled with detailed lectures and step-by-step explanations. It systematically measures an LVLM's capacity to integrate domain-specific multimodal context and execute logical, chain-of-thought reasoning. Performance is quantified using Accuracy, which serves as a robust metric to evaluate the model's problem-solving capabilities and scientific understanding in educational domains.

5.2 Details of Compared Baselines

We evaluate the performance of our proposed SDPR against seven state-of-the-art baselines. These methods include five contrastive decoding-based approaches, which are VCD [14], M3ID [15], ONLY [16], AvisC [17], and MoD [18], as well as two attention intervention-based methods, VAF [1] and VAR [2]. Notably, VAR is designed to mitigate attention sinks by utilizing the RMS norm of hidden states.

We further investigate the empirical failure modes of this hidden state-based selection strategy to demonstrate its limitations in practical scenarios.

The implementation details for these approaches are provided:

(1) **VCD** [14] introduces visual uncertainty by applying Gaussian noise to the input image. It contrasts the original logits $f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t})$ with the distorted logits $f_\theta(y | v', \mathbf{x}, \mathbf{y}_{<t})$ to penalize language priors:

$$p_{vcd}(y_t) = \text{softmax}\left[(1 + \alpha)f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) - \alpha f_\theta(y | v', \mathbf{x}, \mathbf{y}_{<t})\right] \quad (4)$$

where v represents original visual input and v' represents the distorted visual input which is obtained by applying pre-defined distortions (e.g., Gaussian noise mask).

(2) **M3ID** [15] leverages multi-modal mutual information to amplify the conditioning of visual information, especially as the generation length increases and visual signal fades:

$$p_{m3id}(y_t) = \text{softmax}\left[f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) + \frac{1 - e^{-\lambda t}}{e^{-\lambda t}} (f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) - f_\theta(y | \mathbf{x}, \mathbf{y}_{<t}))\right] \quad (5)$$

(3) **ONLY** [16] introduces a training-free, one-layer intervention strategy that dynamically alternates between collaborative and contrastive decoding by monitoring similarity between the original and text-enhanced logit distributions:

$$p_{only}(y_t) = \begin{cases} \text{softmax}\left[f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) + \alpha_1 f_\theta(y | v_{TE}, \mathbf{x}, \mathbf{y}_{<t})\right], & \text{(Collaborative)} \\ \text{softmax}\left[(1 + \alpha_2)f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) - \alpha_2 f_\theta(y | v_{TE}, \mathbf{x}, \mathbf{y}_{<t})\right], & \text{(Contrastive)} \end{cases} \quad (6)$$

where v_{TE} denotes the text-enhanced visual context filtered by the text-to-visual entropy ratio. This dual-path mechanism allows the model to either amplify confident predictions through collaborative decoding when distributions are consistent, or penalize potential hallucinations via contrastive decoding when a significant divergence is detected.

(4) **AvisC** [17] implements attentional vision calibration by identifying and masking "blind" visual tokens that the model ignores or misinterprets during standard inference, leveraging a contrastive mechanism to emphasize grounded visual signals:

$$p_{avisc}(y_t) = \text{softmax}\left[(1 + \alpha)f_\theta(y | v, \mathbf{x}, \mathbf{y}_{<t}) - \alpha f_\theta(y | v^*, \mathbf{x}, \mathbf{y}_{<t})\right] \quad (7)$$

where v^* represents the calibrated visual context with identified blind tokens removed to counteract language-prior-driven overconfidence and ensure more faithful visual grounding.

(5) **MoD** [18] introduces an attention-inspired adaptive strategy that switches between complementary and contrastive decoding based on the consistency of visual attention. Specifically, it measures

the JS divergence between original logits and those derived from attended visual regions:

$$p_{mod}(y_t) = \begin{cases} \text{softmax}\left[f_{\theta}(y | v, \mathbf{x}, \mathbf{y}_{<t}) + \alpha_1 f_{\theta}(y | v_{att}, \mathbf{x}, \mathbf{y}_{<t})\right], & (\text{Collaborative}) \\ \text{softmax}\left[(1 + \alpha_2) f_{\theta}(y | v, \mathbf{x}, \mathbf{y}_{<t}) - \alpha_2 f_{\theta}(y | v_{att}, \mathbf{x}, \mathbf{y}_{<t})\right], & (\text{Contrastive}) \end{cases} \quad (8)$$

Here, v_{att} represents the visual inputs refine by the attention driven from saliency map. This strategy adapts dynamically to amplify reliable visual information or suppress potential hallucinatory noise based on the model’s attentional focus.

(6) **VAF** [1] addresses visual neglect in middle layers by directly amplifying the attention weights of vision-sensitive heads:

$$p_{vaf}(y_t) = f_{\theta}(y | v_{amp}, \mathbf{x}, \mathbf{y}_{<t})$$

where v_{amp} denotes the visually-amplified context. By scaling up attention to visual tokens and suppressing irrelevant prompts, VAF enhances grounding via a single-path intervention without contrastive overhead.

(7) **VAR** [2] identifies and rectifies detrimental attention sinks within the visual-textual attention matrix A by monitoring the RMS norm of hidden states as $RMS(H_j)$, after they zero out identified sink tokens:

$$A_{i,j} = 0, \quad \text{if } RMS(H_j) > \tau$$

where $\tau = 20$. We further analyze the failure of VAR for visual sink localization in Subsection 3.1.

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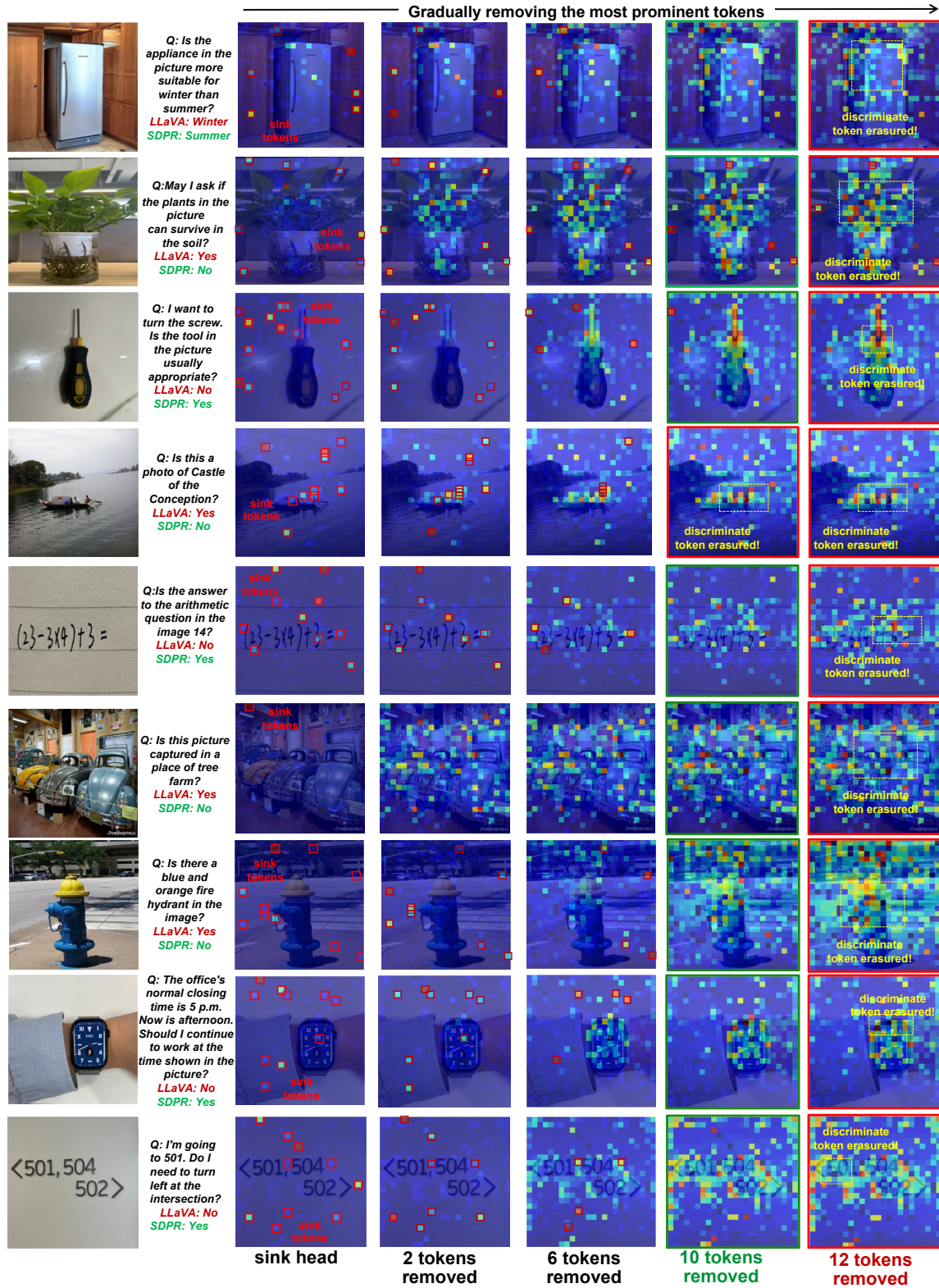


Figure 7: Progressive remove the most significant tokens. Red boxes denote prominent sink tokens. Iterative removal of these tokens releases suppressed visual features, yet removing more than 12 tokens leads to noticeable degradation, as critical information is mistakenly pruned.

Question: Describe this image in detailed.

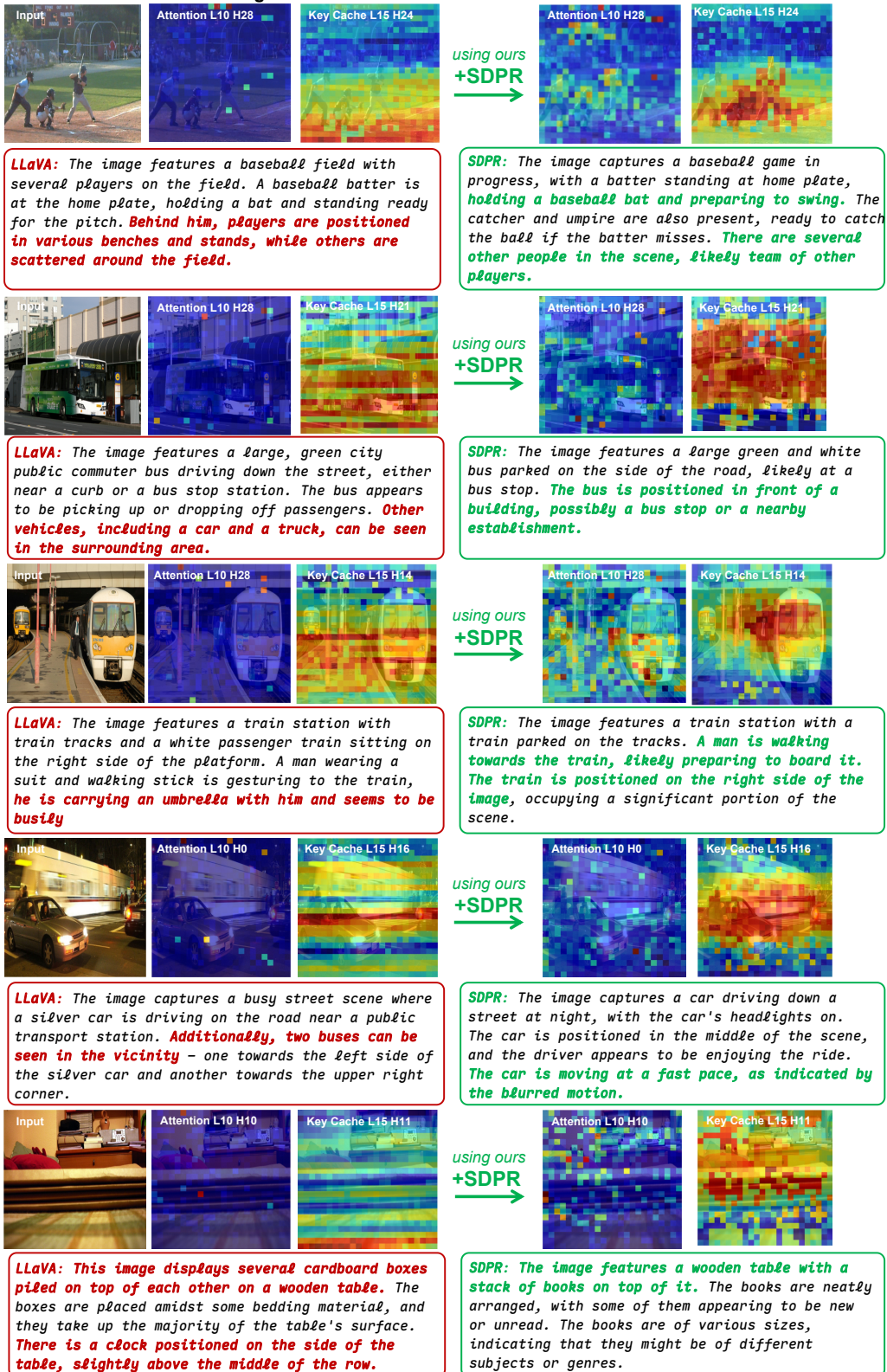


Figure 8: More case studies of regular decoding versus SDPR. The prompt is “Describe this image in detail.” Hallucinated and accurate content are highlighted in red and green.